IT3010

Network Design and Management

Lecture 02

ISO Network Management Framework

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Discover Your Future

IT3010 NETWORK DESIGN MANAGEMENT

Lecture 1

Introduction to Network Design & Management

Today's lecture overview

- Design Methodology and Considerations
- Rationale for Network Management
- Network Management Process
- Network Management Systems

Introduction

Definition of a Data Network: A collection of devices and circuits for transferring data from one computer to another (or device, e.g. printer).

Purpose:

It enables users at different locations to share the resources of a computer stationed elsewhere.

E.g. : Automated Teller Machine (ATM)

Goals

Why bother about the network design..?

Primary goal of network design is to meet the organizations communication needs.

- Productivity ↑
- Budget ↓

Considerations

Achieving the goal..

Need to develop a comprehensive plan. Must take into account the following:

- Suitability
- Reliability
- Scalability
- Durability

Network Engineer/Administrator

[redacted]

Role of a Network Engineer/Administrator

Network engineers have the responsibility for installing, maintaining, troubleshooting, optimizing and expanding the network.

- As a network expands, so too the size and number of potential problems.
- The overall goal of network management is to help network engineers deal with the complexity of data networks.
- Design based on Network Management principles.

The Network Management Process

Network Management is the process of controlling complex data networks to maximize its efficiency and productivity.

ISO Framework for Network Management

- Configuration management
- Security management
- Performance management
- Accounting management
- Fault management

Configuration Management

Configuration management is the process of

- Gathering information about the current network environment.
- Using that data to modify the configuration of network devices.
- Storing the data, maintaining an up-to-date inventory of all network components and producing various reports.

Bridge Configuration Management Information	
Name	Software Version
Payroll Mainframe Subnet	A
Terminal Server Subnet	B
Engineerig Computer subnet	A

Data Collection, Modification and Stori[redacted]

Data Collection

Two methods..

- Manual collection
  - Tedious, error prone, time consuming
- Auto-discovery/Auto-mapping
  - ICMP (ping, traceroute)
  - Network Management Protocol

Data Collection, Modification and Stori[redacted]

Data Modifications

- Once configuration management information has been obtained, it will usually need to be updated.
- Network devices usually contain many pieces of modifiable parameters.
  - E.g. – routers (routing tables, network interfaces), servers (application services, operating systems).

Data Collection, Modification and Stori[redacted]

Storing Information

Methods of storage:

- **Unstructured** (e.g. ASCII files)
  - Advantages:
    - Easily read.
    - Easily accessed from remote locations.
    - Easy to administer.
  - Disadvantages:
    - Inefficient storage.
    - Slow to search.
    - Unable to provide complex data relationships.

Data Collection, Modification and Stori[redacted]

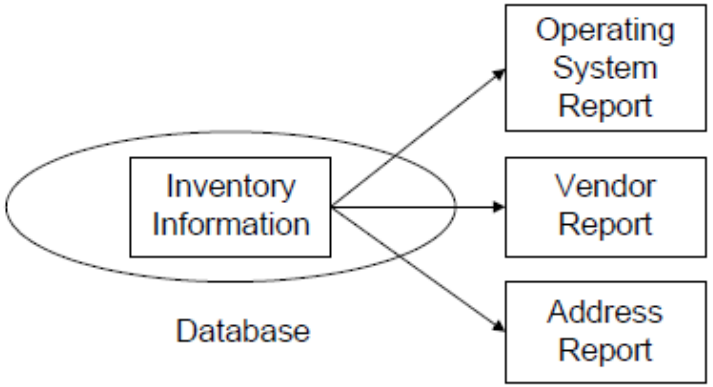
Storing Information

Methods of storage:

- **Structured** (e.g. DBMS)
  - Advantages:
    - Stores data efficiently.
    - Enables users to relate various types of information to one another.
    - Versioning.
  - Disadvantages:
    - Need to learn query language to access data (e.g. SQL).
    - Administrative overheads.

Configuration Management - Benefits

- Automatically gather and update data on network devices.
- Allows devices to be configured remotely.
- Provides central storage location for configuration data.
- Facilitates the production of network inventory and other reports.



Security Management

- Security management involves protecting sensitive information on devices attached to a data network by controlling access points to that information.
- Security management consists of the following aspects:
  - Identifying the sensitive information to be protected.
  - Finding the access points (vulnerabilities).
  - Securing the access points.
  - Maintaining the secure access points.

Security Management - Benefits

- Increases confidence in utilizing the network.
  - A lack of security may force drastic measures, such as eliminating network access of sensitive information altogether.
  - Properly set up and maintained security management can offer more practical alternatives.
- Some examples:
  - 1988: Internet worm
  - More recently: ICMP and TCP-based DoS attacks

Performance Management

Performance management involves ensuring that networks remain accessible and free from congestion:

- Monitoring network devices and their associated links to determine utilization and error rates.
- Helping the network provide consistent quality of service (QoS) by ensuring that the capacity of devices and links is not over taxed to the extent of adversely impacting performance.
  - Context-specific

Performance Management

Performance management entails the following steps:

1. Collecting data on current utilization of network devices and links.
2. Analyzing relevant data.
   - Statistical analysis
   - Workload modeling
3. Setting utilization thresholds.
4. Using simulation to determine how the network can be altered to maximize performance.

Performance Terms

- Availability
- Bandwidth/Throughput
- Propagation
- Congestion
- Latency
- Threshold
- Utilization

Performance Management - Benefits

- Reduces probability of network congestion and inaccessibility so as to provide a consistent level of service to users.
  - E.g. – Knowing the network's utilization workload can help one schedule large data transfers for non-peak times.
- Assist in examining network trends:
  - Content Creators
  - Content-hosting companies
  - Network operators
  - Networking researchers

Accounting Management

Accounting management is the process of gathering network statistics to help the network engineer make decisions about the allocation of network resources.

Accounting management involves the following tasks:

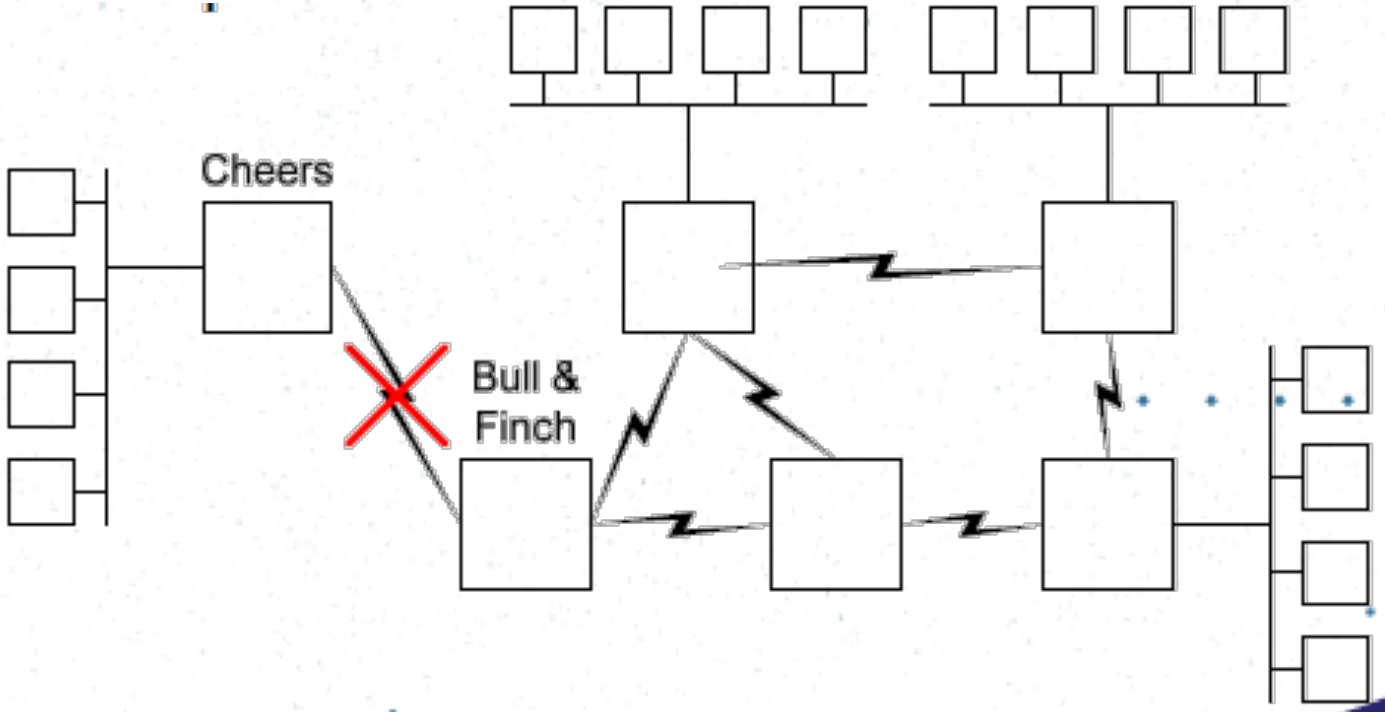
- Gathering data about the utilization of network resources.
  - Need to establish metrics -RFC 1272: "Internet Accounting Background"
    - E.g. Application layer – per-transaction, network layer – number of packets.
- Setting usage quotas using metrics.
- Billing users for their use of the network.

Accounting Management - Benefits

- Allows effective measurement and reporting of accounting information.
- Increases the engineer's understanding of user utilization.
  - Helps the network engineer make informed decisions about the allocation of network resources.
  - Ensure that users have a fair share of the network..

Fault Management

Fault management is the process of locating problems or faults, on the data network.



Fault Management

- The fault management process involves:
  - Detecting symptoms that may lead to a problem/fault
- in the network.
  - Isolating the cause of the symptoms.
    - Find correlations between symptoms and potential problems.
      - Alarms do not usually include explicit information regarding the exact location of the fault.
  - Correcting the problem if possible.

Gathering Information for Fault Management

[redacted]

Two methods..

- Interrupt driven
  - Critical events (e.g. link failure).
  - Solely relying on such events may not facilitate effective fault management.
- Polling
  - Finds faults in a timely manner.
  - Higher bandwidth consumption.
  - Polling can be implemented using ICMP messages (ping).

Fault Management - Benefits

- Enhances network reliability by providing tools to aid/facilitate rapid fault detection, isolation and recovery.
- Maintains the illusion of complete and continuous connectivity between the users and the network.

Network Management Systems

A Network Management System (NMS) comprises:

- An underlying architecture (aka platform).
  - A software package that provides generic/basic functionality of network management for managing a variety of network devices.
- A set of applications built on top of the platform.

Network Management Systems

Example NMSs

- Commercial implementations
  - HP Openview
  - SunConnect SunNet Manager
  - IBM Netview
- Freeware implementations
  - Net-SNMP (formally UCD-SNMP, CMU-SNMP)
  - OpenNMS

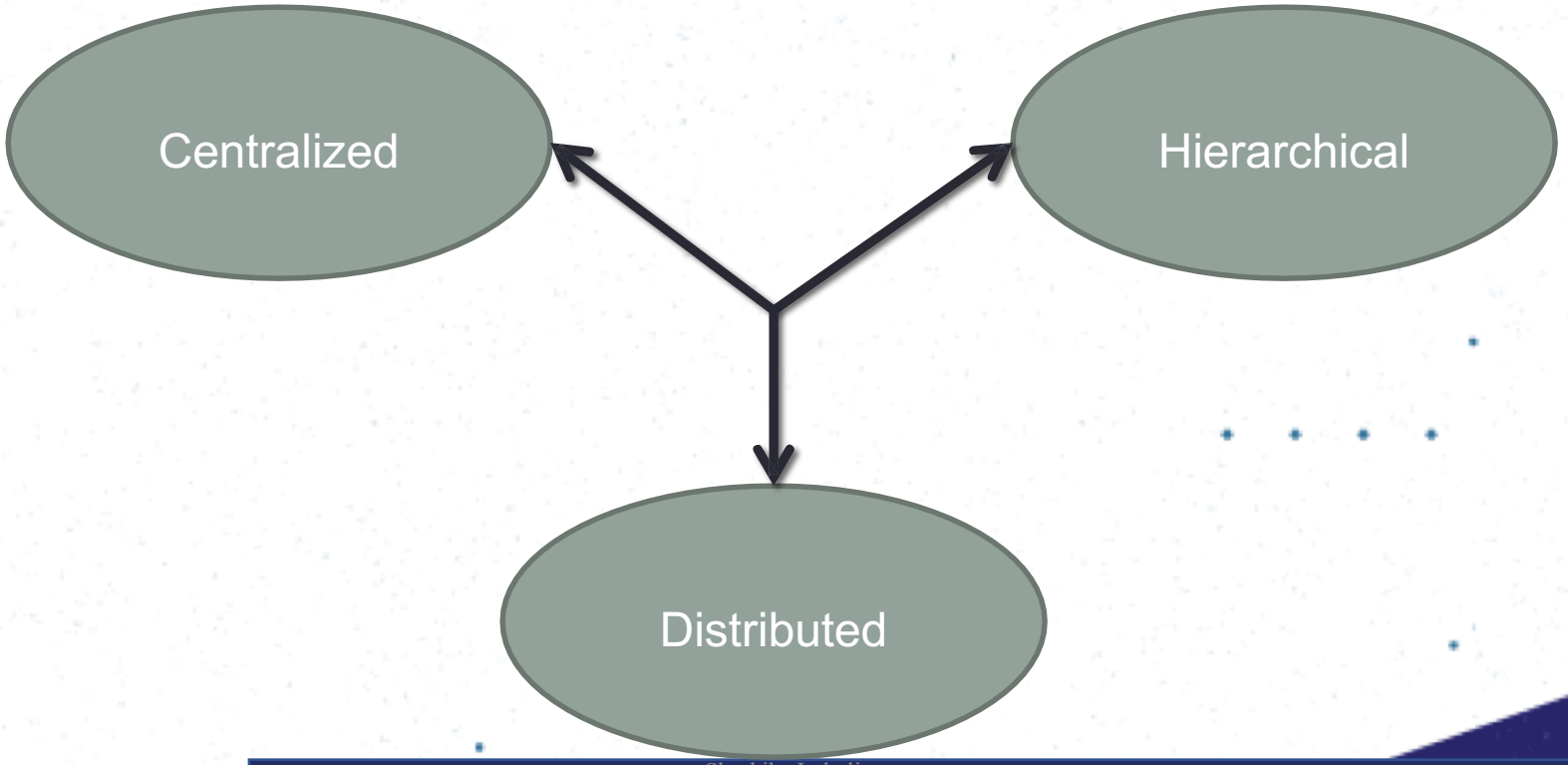
Network Management Systems

The platform should include the following functionality:

- A user interface
- A network map
- A **D**ata**b**ase **M**anagement **S**ystem (DBMS) / **M**anagement **I**nformation **B**ase (MIB)
- A query language
- A customizable menu system
- An event log

Network Management Architecture

An NMS platform can use three architectures to provide functionality.



Next Lecture…!!!

Network Mapping and Baselining

Thank You